

ECE238L Computer Logic Design

Lab 5: Digital Stop Watch

Section 003: Wednesday 2:00-4:45

Name: Tomas Salaz

Frequency Divider VHDL Source Code, Simulation Code, and Constraints

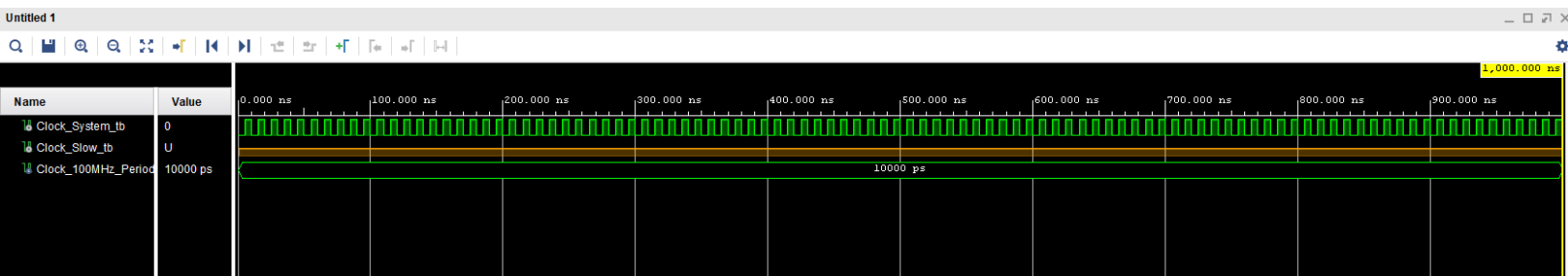
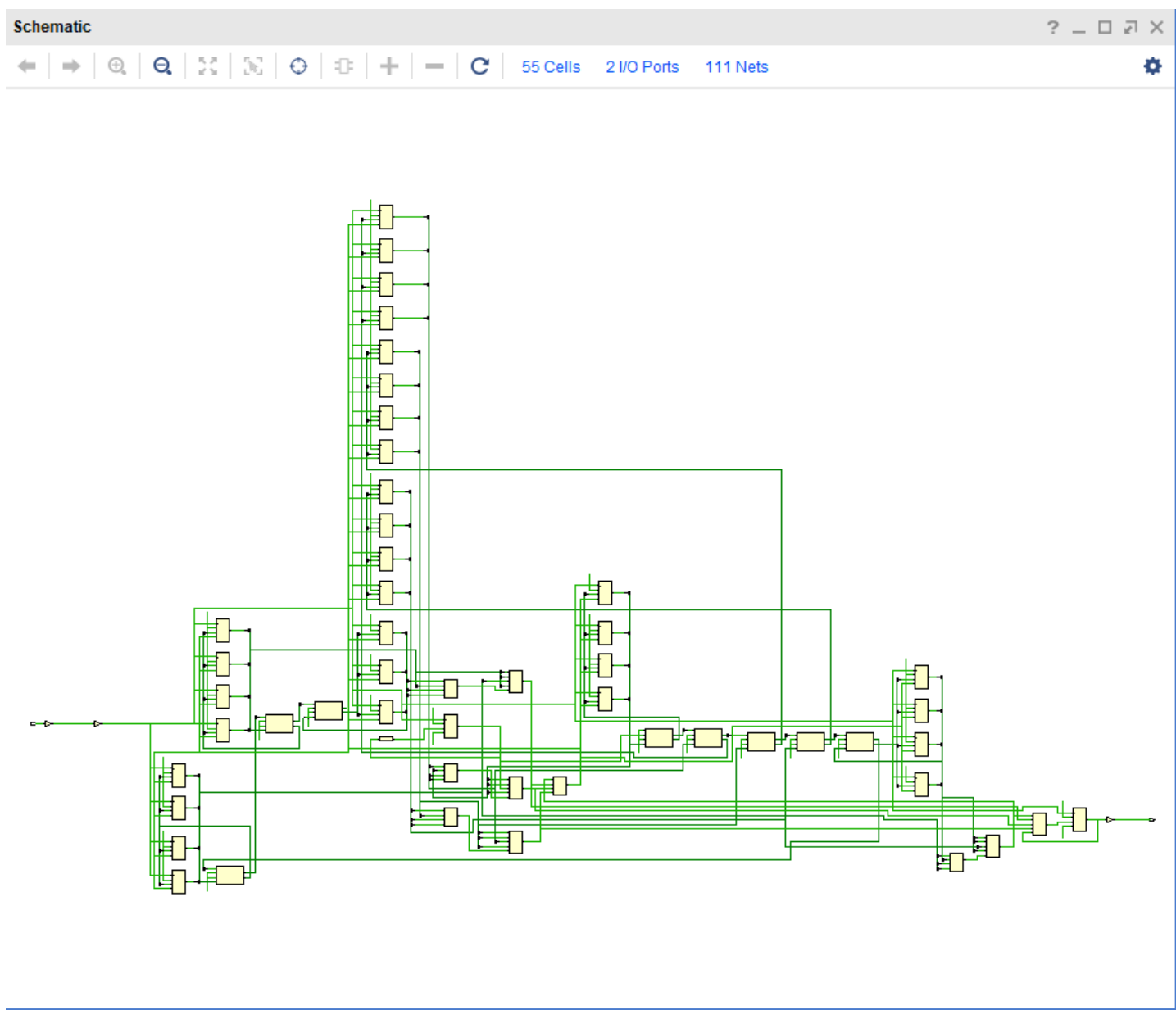
The image displays three VHDL files in a code editor. The top-left window shows the main VHDL source code for the Frequency Divider. The top-right window shows the testbench code for simulating the divider. The bottom window shows the constraints file for the design.

```
FreqDivider.vhd
5 -- Create Date: 03/11/2025 01:26:10 PM
6 -- Design Name:
7 -- Module Name: FreqDivider - Behavioral
8 -- Project Name:
9 -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20
21 library IEEE;
22 use IEEE.STD_LOGIC_1164.ALL;
23 use IEEE.NUMERIC_STD.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity FreqDivider is
35     Port ( Clock_System : in STD_LOGIC;
36           Clock_1Hz : out STD_LOGIC);
37 end FreqDivider;
38
39 architecture Behavioral of FreqDivider is
40     signal count : INTEGER := 1;
41     signal sClock_1Hz : STD_LOGIC := '0';
42
43 begin
44     process (Clock_System)
45     begin
46         if Clock_System' event and Clock_System = '1' then
47             count <= count + 1;
48             if count = 8 then
49                 Clock_1Hz <= not sClock_1Hz;
50                 sClock_1Hz <= not sClock_1Hz;
51                 count <= 1;
52             end if;
53         end if;
54     end process;
55 end Behavioral;
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```

```
FreqDivider_tb.vhd
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24 use IEEE.NUMERIC_STD.ALL;
25
26 -- Uncomment the following library declaration if using
27 -- arithmetic functions with Signed or Unsigned values
28 --use IEEE.NUMERIC_STD.ALL;
29
30 -- Uncomment the following library declaration if instantiating
31 -- any Xilinx leaf cells in this code.
32 --library UNISIM;
33 --use UNISIM.VComponents.all;
34
35 entity FreqDivider_tb is
36     -- Port ( );
37 end FreqDivider_tb;
38
39 architecture Behavioral of FreqDivider_tb is
40     component Frequency_Divider is
41         Port ( Clock_System : in STD_LOGIC;
42               Clock_1Hz : out STD_LOGIC);
43     end component;
44
45     signal Clock_System_tb, Clock_Slow_tb : STD_LOGIC;
46     constant Clock_100MHz_Period : time := 10ns;
47
48     begin
49         uut: Frequency_Divider port map (Clock_System => Clock_System_tb,
50                                           Clock_1Hz => Clock_Slow_tb);
51
52         Clk_process : process
53         begin
54             Clock_System_tb <= '0'; wait for Clock_100MHz_Period/2;
55             Clock_System_tb <= '1'; wait for Clock_100MHz_Period/2;
56         end process;
57     end Behavioral;
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```

```
FreqDivider_constr.xdc
1 set_property IOSTANDARD LVCMOS33 [get_ports Clock_1Hz]
2 set_property IOSTANDARD LVCMOS33 [get_ports Clock_System]
3 set_property PACKAGE_PIN E3 [get_ports Clock_System]
4 set_property PACKAGE_PIN H17 [get_ports Clock_1Hz]
5
```

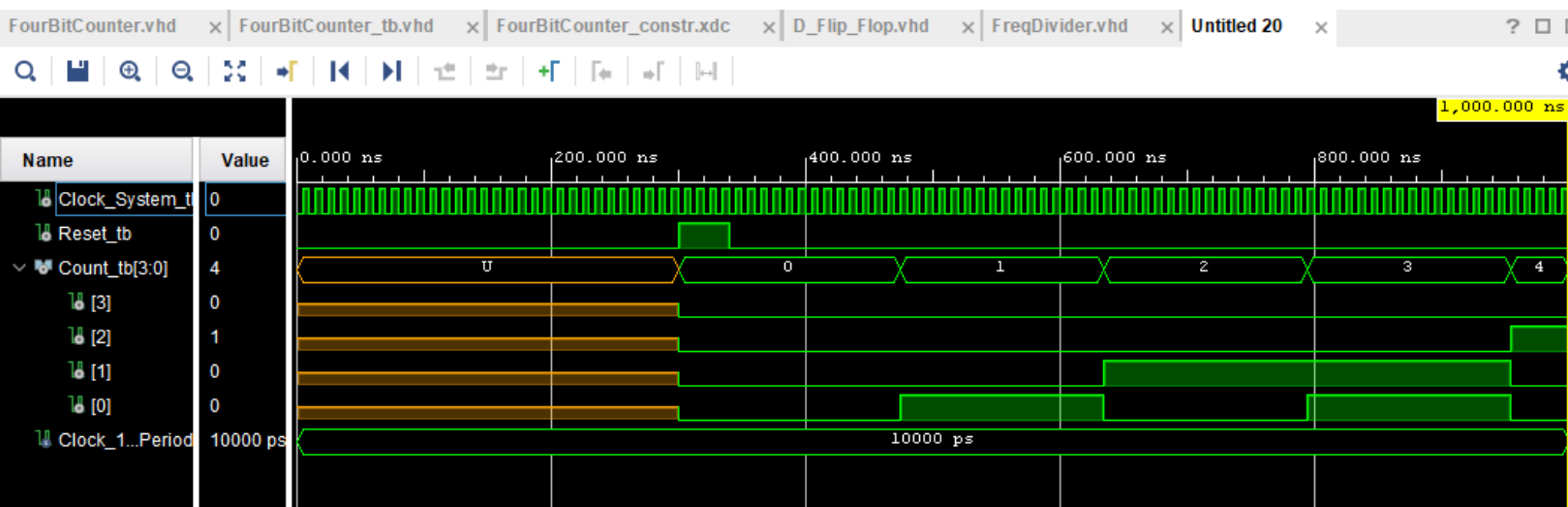
Frequency Divider Waveform and Schematic



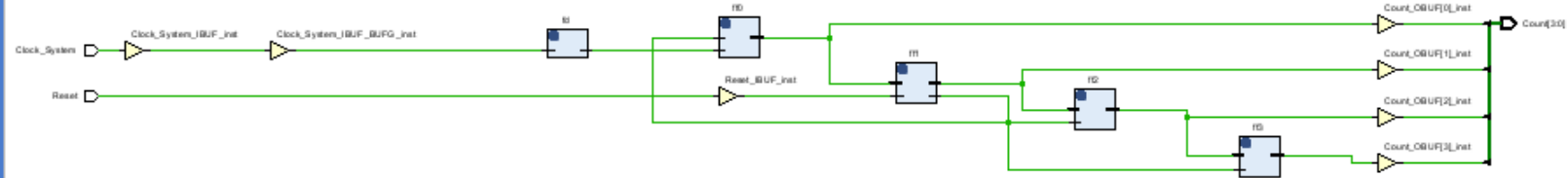
Counter VHDL Source Code, Simulation Code, and Constraints

```
FourBitCounter.vhd
C:/Users/salaz25/ECE238L/FourBitCounter/FourBitCounter.srcs/sources_1/new/FourBitCounter.vhd
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20
21 library IEEE;
22 use IEEE.STD_LOGIC_1164.ALL;
23
24 -- Uncomment the following library declaration if using
25 -- arithmetic functions with Signed or Unsigned values
26 --use IEEE.NUMERIC_STD.ALL;
27
28 -- Uncomment the following library declaration if instantiating
29 -- any Xilinx leaf cells in this code.
30 --library UNISIM;
31 --use UNISIM.VComponents.all;
32
33 entity FourBitCounter is
34     Port ( Clock_System : in STD_LOGIC;
35           Reset : in STD_LOGIC;
36           Count : out STD_LOGIC_VECTOR (3 downto 0));
37 end FourBitCounter;
38
39 architecture Behavioral of FourBitCounter is
40
41     component FreqDivider is
42         Port ( Clock_System : in STD_LOGIC;
43               Clock_1Hz : out STD_LOGIC);
44     end component;
45
46     component D_Flip_Flop is
47         Port ( D : in STD_LOGIC;
48               CLK, CLR : in STD_LOGIC;
49               Q : out STD_LOGIC);
50     end component;
51
52     signal temp : STD_LOGIC_VECTOR (3 downto 0);
53     signal notTemp : STD_LOGIC_VECTOR (3 downto 0);
54     signal Clk_1Hz : STD_LOGIC := '0';
55
56 begin
57     notTemp <= not (temp);
58     fd : FreqDivider port map (Clock_System => Clock_System, Clock_1Hz => Clk_1Hz);
59     ff0 : D_Flip_Flop port map (CLK => Clk_1Hz, D => notTemp(0), Q => temp(0), CLR => reset);
60     ff1 : D_Flip_Flop port map (CLK => temp(0), D => notTemp(1), Q => temp(1), CLR => reset);
61     ff2 : D_Flip_Flop port map (CLK => temp(1), D => notTemp(2), Q => temp(2), CLR => reset);
62     ff3 : D_Flip_Flop port map (CLK => temp(2), D => notTemp(3), Q => temp(3), CLR => reset);
63     Count <= temp;
64
65 end Behavioral;
66
67
FourBitCounter_tb.vhd
14
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20
21 library IEEE;
22 use IEEE.STD_LOGIC_1164.ALL;
23
24 -- Uncomment the following library declaration if using
25 -- arithmetic functions with Signed or Unsigned values
26 --use IEEE.NUMERIC_STD.ALL;
27
28 -- Uncomment the following library declaration if instantiating
29 -- any Xilinx leaf cells in this code.
30 --library UNISIM;
31 --use UNISIM.VComponents.all;
32
33 entity FourBitCounter_tb is
34     Port ( );
35 end FourBitCounter_tb;
36
37 architecture Behavioral of FourBitCounter_tb is
38
39     Component FourBitCounter is
40         Port ( Clock_System : in STD_LOGIC;
41               Reset : in STD_LOGIC;
42               Count : out STD_LOGIC_VECTOR (3 downto 0));
43     end component;
44
45     signal Clock_System_tb, Reset_tb : STD_LOGIC; -- Internal test
46     signal Count_tb : STD_LOGIC_VECTOR (3 downto 0);
47     constant Clock_100MHz_Period : time := 10ns; -- Clock period fo
48
49     begin
50         uut : FourBitCounter port map (Clock_System => Clock_System_tb,
51                                       Reset => Reset_tb,
52                                       Count => Count_tb);
53
54         Clk_process: process
55         begin
56             Clock_System_tb <= '0'; wait for Clock_100MHz_Period/2;
57             Clock_System_tb <= '1'; wait for Clock_100MHz_Period/2;
58         end process;
59
60         stimulus: process
61         begin
62             Reset_tb <= '0'; wait for 300 ns;
63             Reset_tb <= '1'; wait for 40 ns;
64             Reset_tb <= '0'; wait for 600 ns;
65         end process;
66     end Behavioral;
67
68
FourBitCounter_constr.xdc
1 set_property IOSTANDARD LVCMOS33 [get_ports {Count[3]}]
2 set_property IOSTANDARD LVCMOS33 [get_ports {Count[2]}]
3 set_property IOSTANDARD LVCMOS33 [get_ports {Count[1]}]
4 set_property IOSTANDARD LVCMOS33 [get_ports {Count[0]}]
5 set_property IOSTANDARD LVCMOS33 [get_ports {Clock_System}]
6 set_property PACKAGE_PIN J15 [get_ports {Reset}]
7 set_property PACKAGE_PIN E3 [get_ports {Clock_System}]
8 set_property PACKAGE_PIN J15 [get_ports {Reset}]
9 set_property PACKAGE_PIN N14 [get_ports {Count[3]}]
10 set_property PACKAGE_PIN J13 [get_ports {Count[2]}]
11 set_property PACKAGE_PIN K15 [get_ports {Count[1]}]
12 set_property PACKAGE_PIN H17 [get_ports {Count[0]}]
13
```

Counter Waveform



Counter Schematic



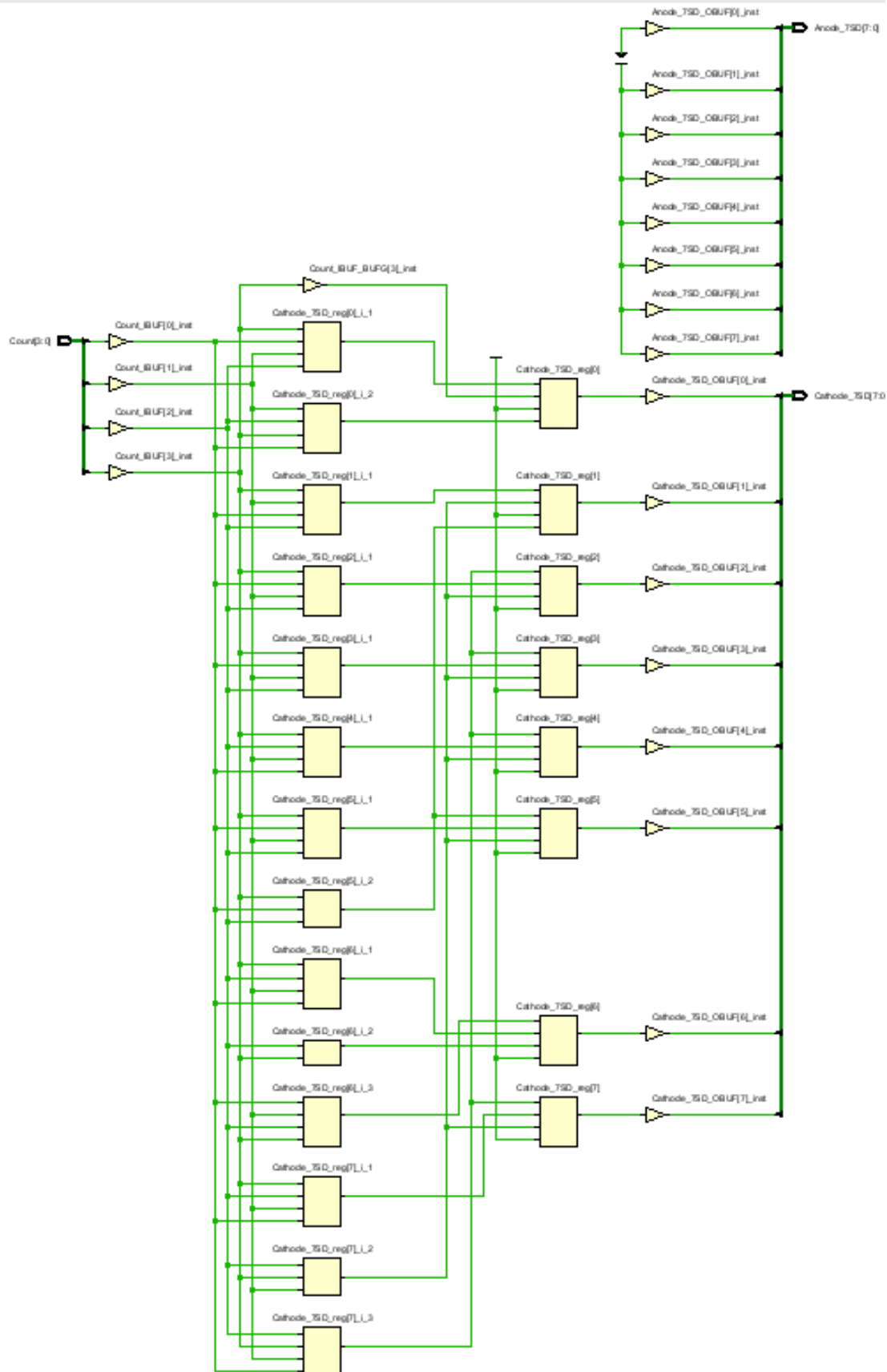
Display VHDL Source Code, Simulation Code, and Constraints

```
DisplayDigits.vhd
13  -- Dependencies:
14  --
15  -- Revision:
16  -- Revision 0.01 - File Created
17  -- Additional Comments:
18  --
19  --
20  --
21  --
22  library IEEE;
23  use IEEE.STD_LOGIC_1164.ALL;
24  --
25  -- Uncomment the following library declaration if using
26  -- arithmetic functions with Signed or Unsigned values
27  --use IEEE.NUMERIC_STD.ALL;
28  --
29  -- Uncomment the following library declaration if instantiating
30  -- any Xilinx leaf cells in this code.
31  --library UNISIM;
32  --use UNISIM.VComponents.all;
33  --
34  entity DisplayDigits is
35  Port ( Count : in STD_LOGIC_VECTOR (3 downto 0);
36        Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
37        Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
38  end DisplayDigits;
39  --
40  architecture Behavioral of DisplayDigits is
41  begin
42  process (Count)
43  begin
44  if Count = "0000" then Cathode_7SD <= "00000011"; --0
45  elsif Count = "0010" then Cathode_7SD <= "10011111";
46  elsif Count = "0010" then Cathode_7SD <= "00100101";
47  elsif Count = "0011" then Cathode_7SD <= "00001101";
48  elsif Count = "0100" then Cathode_7SD <= "10011001";
49  elsif Count = "0101" then Cathode_7SD <= "01001001";
50  elsif Count = "0110" then Cathode_7SD <= "11000001";
51  elsif Count = "0111" then Cathode_7SD <= "00011111";
52  elsif Count = "1000" then Cathode_7SD <= "00000001";
53  elsif Count = "1001" then Cathode_7SD <= "00001001"; --9
54  --
55  elsif Count = "1010" then Cathode_7SD <= "00010001"; --A
56  elsif Count = "1011" then Cathode_7SD <= "11000001";
57  elsif Count = "1100" then Cathode_7SD <= "01100011";
58  elsif Count = "1101" then Cathode_7SD <= "10000101";
59  elsif Count = "1110" then Cathode_7SD <= "01100001";
60  elsif Count = "1111" then Cathode_7SD <= "01110001"; --F
61  end if;
62  Anode_7SD <= "11111110";
63  end process;
64  end Behavioral;
65  --

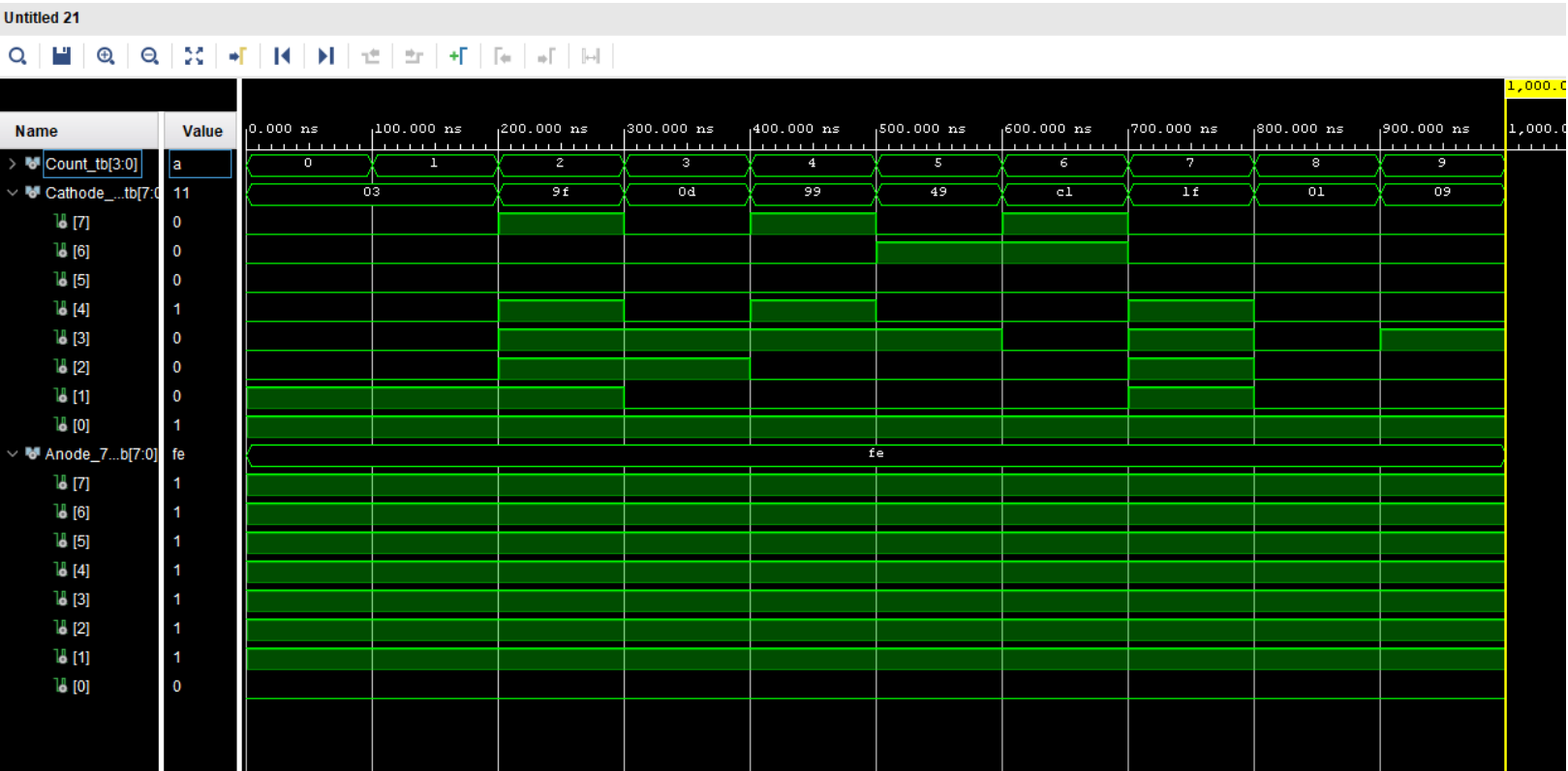
DisplayDigits_tb.vhd
23  use IEEE.STD_LOGIC_1164.ALL;
24  --
25  -- Uncomment the following library declaration if using
26  -- arithmetic functions with Signed or Unsigned values
27  --use IEEE.NUMERIC_STD.ALL;
28  --
29  -- Uncomment the following library declaration if instantiating
30  -- any Xilinx leaf cells in this code.
31  --library UNISIM;
32  --use UNISIM.VComponents.all;
33  --
34  entity DisplayDigits_tb is
35  Port ( );
36  end DisplayDigits_tb;
37  --
38  architecture bench of DisplayDigits_tb is
39  component DisplayDigits is
40  Port ( Count : in STD_LOGIC_VECTOR (3 downto 0);
41        Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
42        Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
43  end component;
44  --
45  signal Count_tb : STD_LOGIC_VECTOR (3 downto 0);
46  signal Cathode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
47  signal Anode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
48  --
49  begin
50  uut : DisplayDigits port map (Count => Count_tb,
51                                Cathode_7SD => Cathode_7SD_tb,
52                                Anode_7SD => Anode_7SD_tb);
53  --
54  stimulus : process
55  begin
56  Count_tb <= "0000"; wait for 100 ns;
57  Count_tb <= "0001"; wait for 100 ns;
58  Count_tb <= "0010"; wait for 100 ns;
59  Count_tb <= "0011"; wait for 100 ns;
60  Count_tb <= "0100"; wait for 100 ns;
61  Count_tb <= "0101"; wait for 100 ns;
62  Count_tb <= "0110"; wait for 100 ns;
63  Count_tb <= "0111"; wait for 100 ns;
64  Count_tb <= "1000"; wait for 100 ns;
65  Count_tb <= "1001"; wait for 100 ns;
66  Count_tb <= "1010"; wait for 100 ns;
67  Count_tb <= "1011"; wait for 100 ns;
68  Count_tb <= "1100"; wait for 100 ns;
69  Count_tb <= "1101"; wait for 100 ns;
70  Count_tb <= "1110"; wait for 100 ns;
71  Count_tb <= "1111"; wait for 100 ns;
72  wait;
73  end process;
74  end bench;
75  --

DisplayDigits_constr.xdc
1  set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[7]}]
2  set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[6]}]
3  set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[5]}]
4  set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[4]}]
5  set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[3]}]
6  set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[2]}]
7  set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[1]}]
8  set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[0]}]
9  set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[7]}]
10 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[6]}]
11 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[5]}]
12 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[4]}]
13 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[3]}]
14 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[2]}]
15 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[1]}]
16 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[0]}]
17 set_property IOSTANDARD LVCMOS33 [get_ports {Count[3]}]
18 set_property IOSTANDARD LVCMOS33 [get_ports {Count[2]}]
19 set_property IOSTANDARD LVCMOS33 [get_ports {Count[1]}]
20 set_property IOSTANDARD LVCMOS33 [get_ports {Count[0]}]
21 set_property PACKAGE_PIN J15 [get_ports {Count[0]}]
22 set_property PACKAGE_PIN L16 [get_ports {Count[1]}]
23 set_property PACKAGE_PIN M13 [get_ports {Count[2]}]
24 set_property PACKAGE_PIN R15 [get_ports {Count[3]}]
25 set_property PACKAGE_PIN H15 [get_ports {Cathode_7SD[0]}]
26 set_property PACKAGE_PIN L18 [get_ports {Cathode_7SD[1]}]
27 set_property PACKAGE_PIN T11 [get_ports {Cathode_7SD[2]}]
28 set_property PACKAGE_PIN P15 [get_ports {Cathode_7SD[3]}]
29 set_property PACKAGE_PIN K13 [get_ports {Cathode_7SD[4]}]
30 set_property PACKAGE_PIN K16 [get_ports {Cathode_7SD[5]}]
31 set_property PACKAGE_PIN R10 [get_ports {Cathode_7SD[6]}]
32 set_property PACKAGE_PIN T10 [get_ports {Cathode_7SD[7]}]
33 set_property PACKAGE_PIN J17 [get_ports {Anode_7SD[0]}]
34 set_property PACKAGE_PIN J18 [get_ports {Anode_7SD[1]}]
35 set_property PACKAGE_PIN T9 [get_ports {Anode_7SD[2]}]
36 set_property PACKAGE_PIN J14 [get_ports {Anode_7SD[3]}]
37 set_property PACKAGE_PIN P14 [get_ports {Anode_7SD[4]}]
38 set_property PACKAGE_PIN T14 [get_ports {Anode_7SD[5]}]
39 set_property PACKAGE_PIN K2 [get_ports {Anode_7SD[6]}]
40 set_property PACKAGE_PIN U13 [get_ports {Anode_7SD[7]}]
41
```

Display Waveform



Display Waveform



Updated D-Flip-Flop Code

```

41
42 library IEEE;
43 use IEEE.STD_LOGIC_1164.ALL;
44
45 -- Uncomment the following library declaration if using
46 -- arithmetic functions with Signed or Unsigned values
47 --use IEEE.NUMERIC_STD.ALL;
48
49 -- Uncomment the following library declaration if instantiating
50 -- any Xilinx leaf cells in this code.
51 --library UNISIM;
52 --use UNISIM.VComponents.all;
53
54 entity D_Flip_Flop is
55     Port ( D : in STD_LOGIC;
56           CLK, CLR : in STD_LOGIC;
57           Q : out STD_LOGIC);
58 end D_Flip_Flop;
59
60 architecture Behavioral of D_Flip_Flop is
61     begin
62         state : process (CLK, CLR)
63         begin
64             if CLR = '1' then
65                 Q <= '0';
66             elsif falling_edge (CLK) then
67                 Q <= D;
68             end if;
69         end process;
70     end Behavioral;
71
72 end

```

Top Module VHDL Source Code

```
Lab5TopMod.vhd
C:/Users/tsalaz25/ECE238L/Lab5TopMod/Lab5TopMod.srscs/sources_1/new/Lab5TopMod.vhd

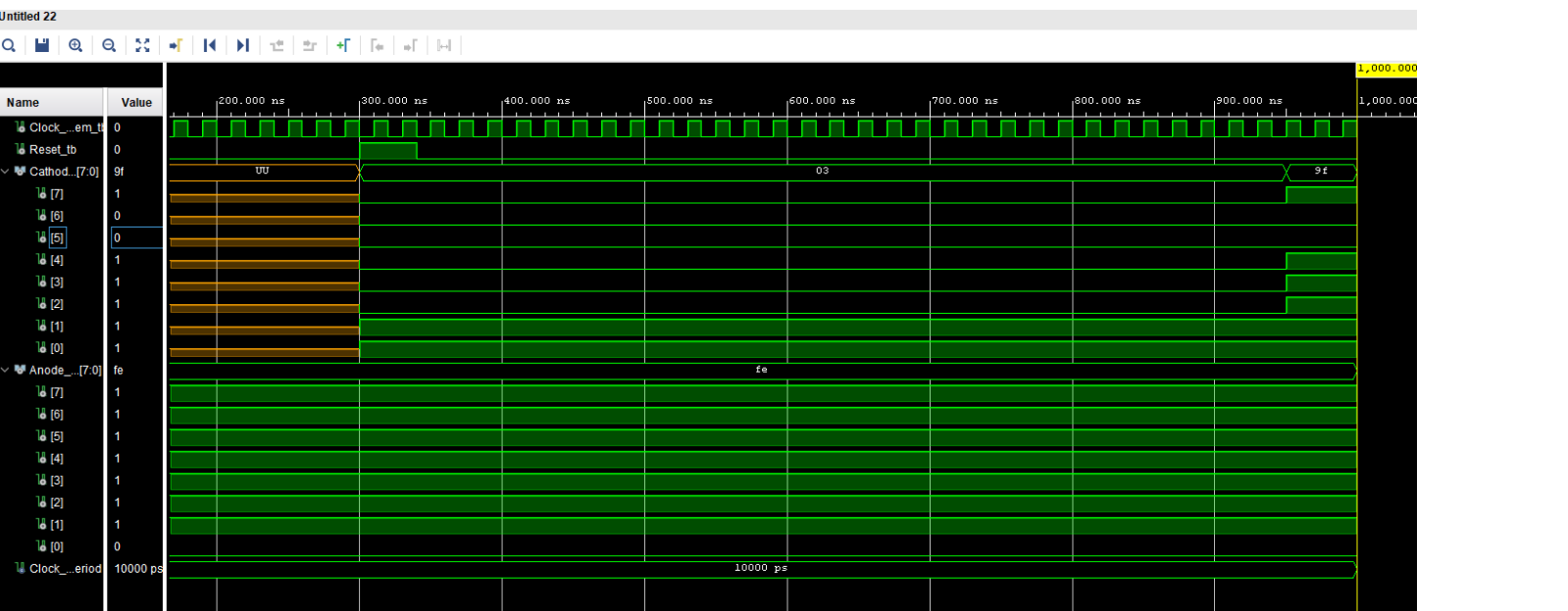
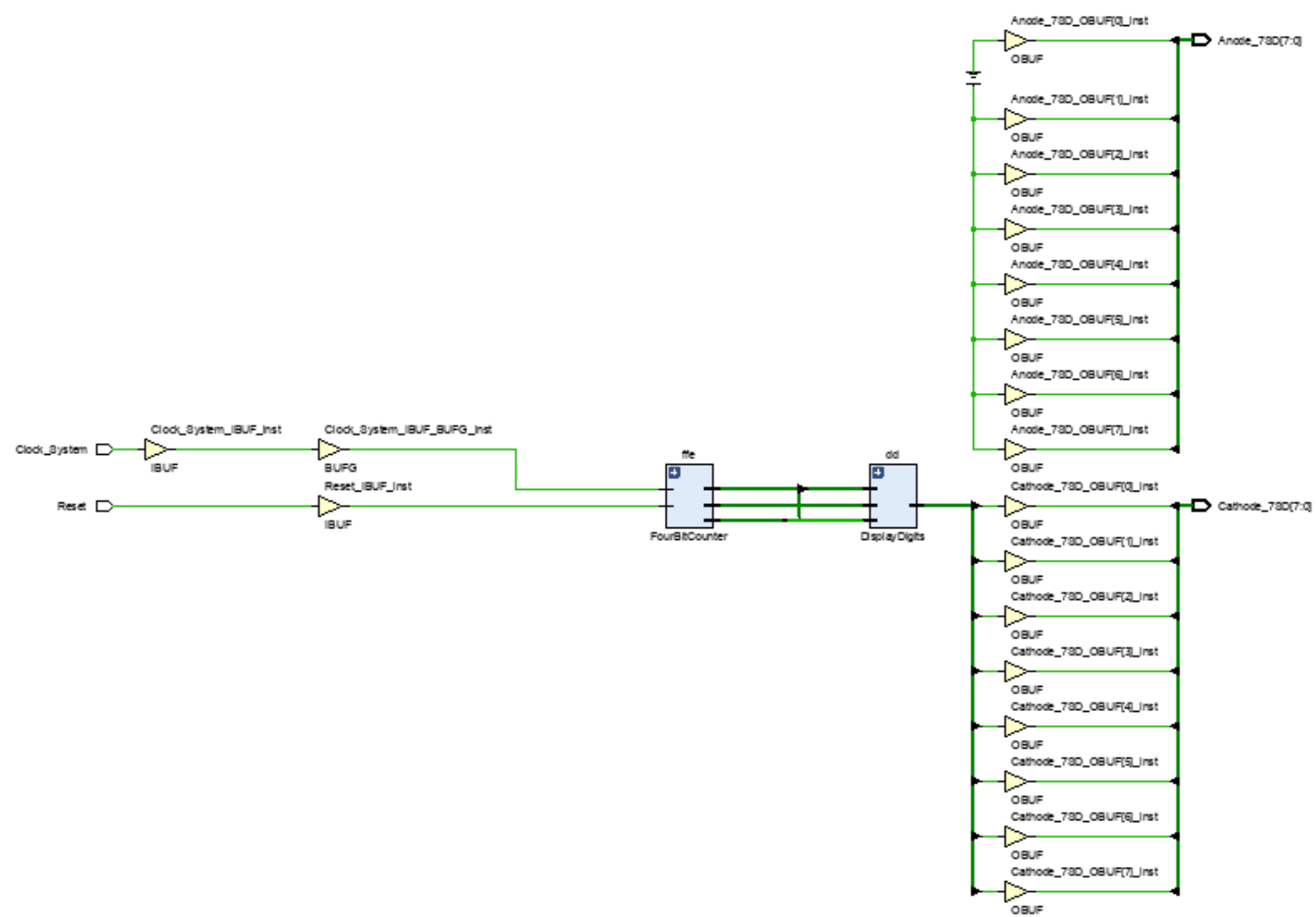
10  -- Tool Versions:
11  -- Description:
12  --
13  -- Dependencies:
14  --
15  -- Revision:
16  -- Revision 0.01 - File Created
17  -- Additional Comments:
18  --
19  -----
20
21
22  library IEEE;
23  use IEEE.STD_LOGIC_1164.ALL;
24
25  -- Uncomment the following library declaration if using
26  -- arithmetic functions with Signed or Unsigned values
27  --use IEEE.NUMERIC_STD.ALL;
28
29  -- Uncomment the following library declaration if instantiating
30  -- any Xilinx leaf cells in this code.
31  --library UNISIM;
32  --use UNISIM.VComponents.all;
33
34  entity Lab5TopMod is
35      Port ( Clock_System : in STD_LOGIC;
36            Reset : in STD_LOGIC;
37            Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
38            Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
39  end Lab5TopMod;
40
41  architecture Behavioral of Lab5TopMod is
42
43      component FourBitCounter is
44          Port ( Clock_System : in STD_LOGIC;
45                Reset : in STD_LOGIC;
46                Count : out STD_LOGIC_VECTOR (3 downto 0));
47      end component;
48
49      component DisplayDigits is
50          Port ( Count : in STD_LOGIC_VECTOR (3 downto 0);
51                Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
52                Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
53      end component;
54
55      signal COUNT : STD_LOGIC_VECTOR (3 downto 0);
56
57      begin
58          ffe : FourBitCounter port map (Clock_System => Clock_System, Reset => Reset, Count => COUNT);
59          dd : DisplayDigits port map (count => COUNT, Cathode_7SD => Cathode_7SD, Anode_7SD => Anode_7SD);
60
61      end Behavioral;
62
```

Top Module Simulation Code and Constraints

```
Lab5TopMod_tb.vhd
C:/Users/tsalaz25/ECE238L/Lab5TopMod/Lab5TopMod.srscs/sim_1/new/Lab5TopMod_tb.vhd

20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity Lab5TopMod_tb is
35     -- Port ( );
36 end Lab5TopMod_tb;
37
38 architecture bench of Lab5TopMod_tb is
39
40     component Lab5TopMod
41         Port ( Clock_System : in STD_LOGIC;
42               Reset : in STD_LOGIC;
43               Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
44               Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
45     end component;
46
47     signal Clock_System_tb : STD_LOGIC;
48     signal Reset_tb : STD_LOGIC;
49     signal Cathode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
50     signal Anode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
51     constant Clock_100MHz_period : time := 10ns;
52     begin
53         uut : Lab5TopMod port map (Clock_System => Clock_System_tb,
54                                   Reset => Reset_tb,
55                                   Cathode_7SD => Cathode_7SD_tb,
56                                   Anode_7SD => Anode_7SD_tb);
57
58         CLK_process : process
59         begin
60             Clock_System_tb <= '0' ; wait for Clock_100MHz_period;
61             Clock_System_tb <= '1' ; wait for Clock_100MHz_period;
62         end process;
63
64         stimulus : process
65         begin
66             Reset_tb <= '0'; wait for 300 ns;
67             Reset_tb <= '1'; wait for 40 ns;
68             Reset_tb <= '0'; wait for 600 ns;
69         end process;
70
71     end bench;
72
```

Top Module Schematic and Waveform



Summary:

The goal for this lab was simple but the implementation was challenging. The first thing we did was make the Frequency Divider which was used implement some of the counting process for the Atrix-7. This was done by counting the rising edge of the system clock. Next, we had to implement the Counter itself, which had 2 components, The Frequency Divider and the Updated D-Flip-Flop, which was used for counting every time the number was incremented and flipping back to 0 when the 4-Bit number was maxed out. The last element was the 7-Segment display, which was implemented easily since we have already worked with the Display. This time it was implemented to display the number on the counter. The last part was making a Top Module and using all these components to make a Digital Stop Watch.

Problems:

The only problems I had were with the Counter implementation and fixing my D-Flip-Flop to be compatible with this Code. Ended up being an easy fix since I had to add a few lines of code, but making it work within the counter did give me some trouble.

Hints:

Make sure you know what everything is doing, or have a pretty good idea. The later parts of this lab get a bit messy if you don't have a sense of what is going on.

Recommendations:

None, I just a really long lab. Maybe find a way to shorten it.